

## Star Vetter - confirmed eclipsing binary candidates

| TIC       | Period (d) | Depth   | BLS SNR | Pixel SNR | Offset | Tier        |
|-----------|------------|---------|---------|-----------|--------|-------------|
| 437150434 | 5.2853     | 15.003% | 1081    | 85.33     | 1"     | Clean       |
| 237115819 | 14.7921    | 8.883%  | 507     | 27.82     | 4"     | Clean       |
| 66732476  | 3.1087     | 9.814%  | 370     | 26.93     | 8"     | Clean       |
| 29495157  | 5.1474     | 8.579%  | 333     | 52.62     | 5"     | Alias       |
| 119459681 | 5.9442     | 15.977% | 311     | 68.15     | 2"     | Clean       |
| 411434067 | 1.3536     | 4.289%  | 193     | 33.70     | 4"     | Alias       |
| 330901906 | 3.7321     | 0.498%  | 60      | 87.24     | 4"     | Thin margin |
| 303470667 | 5.4319     | 2.220%  | 57      | 5.02      | 9"     | Marginal    |
| 273158289 | 1.1519     | 1.255%  | 41      | 17.24     | 2"     | Thin margin |
| 111149941 | 5.2401     | 0.045%  | 10      | 9.73      | 21"    | Ambiguous   |
| 251898081 | 27.3089    | 18.630% | 5535    | 464.16    | 3"     | Alias       |
| 467214801 | 0.7609     | 0.950%  | 2134    | 21.36     | 0"     | Alias       |
| 399770041 | 1.5802     | 4.291%  | 489     | 36.92     | 1"     | Clean       |
| 349701853 | 30.3427    | 28.445% | 343     | 51.05     | 6"     | Clean       |
| 233459038 | 25.3384    | 8.315%  | 237     | 84.46     | 5"     | Alias       |
| 281310348 | 1.7702     | 7.273%  | 180     | 57.87     | 2"     | Clean       |
| 318887646 | 14.2817    | 3.605%  | 150     | 6.94      | 13"    | Clean       |
| 444465823 | 18.5133    | 3.476%  | 147     | 25.18     | 1"     | Alias       |
| 380943251 | 0.7209     | 7.667%  | 101     | 15.26     | 3"     | Clean       |
| 66716482  | 8.0283     | 3.329%  | 65      | 21.51     | 7"     | Clean       |
| 309656476 | 28.9204    | 0.050%  | 25      | 13.80     | 14"    | Clean       |
| 229494833 | 15.3839    | 0.037%  | 15      | 5.19      | 12"    | Alias       |
| 44533527  | 14.5952    | 0.101%  | 12      | 7.33      | 16"    | Clean       |
| 220423668 | 3.6511     | 4.525%  | 755     | 17.67     | 6"     | Alias       |
| 365998904 | 1.2494     | 2.398%  | 375     | 15.52     | 3"     | Clean       |
| 233452636 | 1.9063     | 5.467%  | 202     | 8.94      | 2"     | Thin margin |
| 437171483 | 1.4735     | 4.947%  | 923     | 73.37     | 3"     | Clean       |
| 411457490 | 2.4324     | 5.531%  | 431     | 58.00     | 1"     | Alias       |
| 295373509 | 5.4219     | 8.026%  | 247     | 22.12     | 2"     | Clean       |
| 374158907 | 14.6563    | 4.974%  | 179     | 21.40     | 2"     | Alias       |

# TIC 59490510

Eclipsing binary candidate dossier - Star Vetter

CLEAN

## BLS EPHEMERIS

Period 9.124822 d  
Epoch (t0) 2503.5152 BTJD  
Duration 0.1200 d  
Depth 32.990 % (329,905 ppm)  
SNR 1068.4

## CATALOG CROSS-MATCH (novelty)

|                              |          |
|------------------------------|----------|
| VSX                          | no match |
| ASAS-SN                      | no match |
| Gaia eclipsing-binary table  | no match |
| Gaia variable classification | no match |
| TESS EB catalog (Prsa+2022)  | no match |
| ExoFOP TOI list              | no match |

## INDEPENDENT CHECKS (f3i gates)

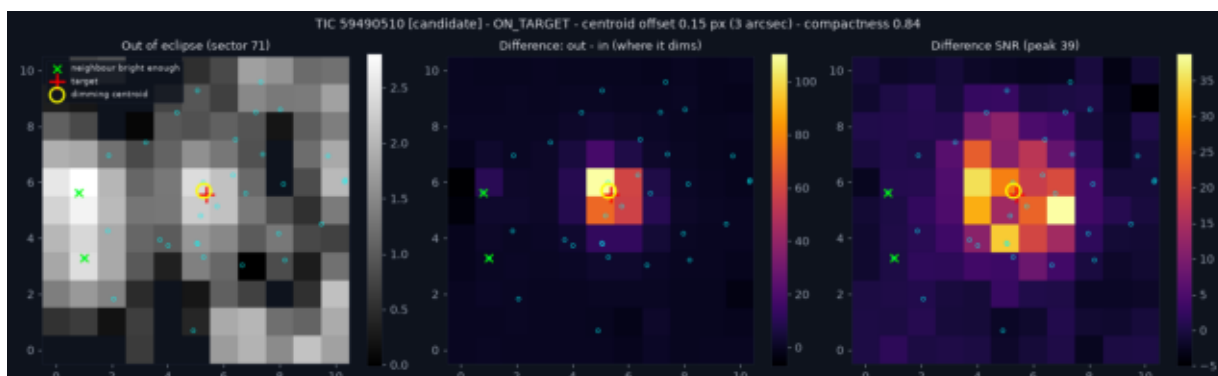
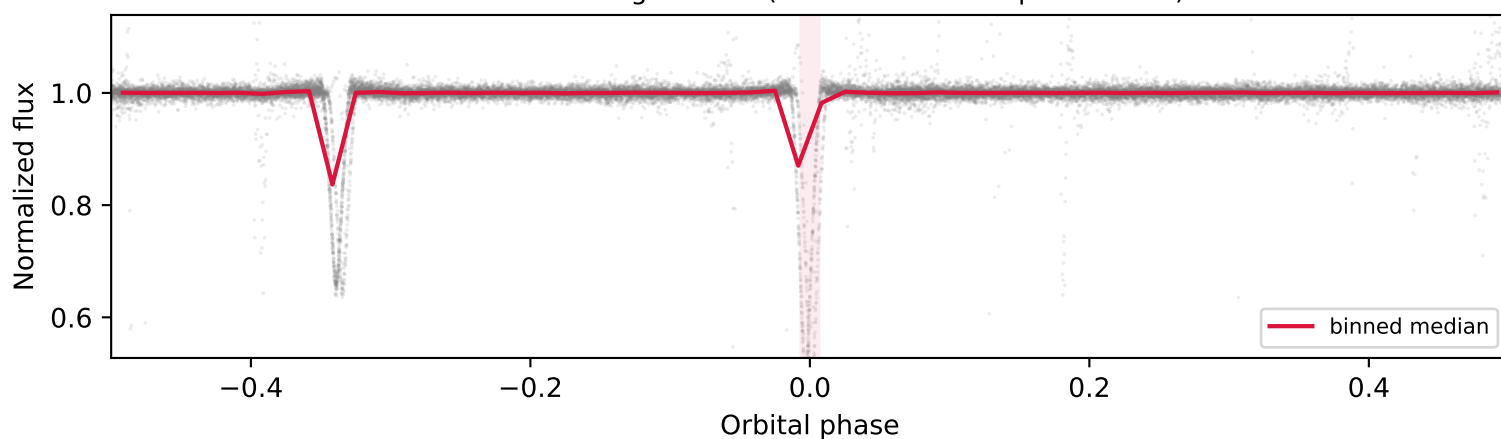
Eclipses observed 12 (gate  $\geq 3$ )  
Robust depth ratio 1.06 (gate  $\geq 0.50$ )  
Robust depth sigma 761.9 (gate  $\geq 7.0$ )  
Odd/even z-score 2.65 (consistent)  
Secondary eclipse detected, phase -0.33, 9.7 sigma

Verdict: NOVEL

## PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 71 light-curve dip in this sector: 310.7 sigma t0 refined by -0.0050 phase  
Difference-image peak SNR 38.74 (gate  $\geq 5.0$ ) Compactness 0.84 (gate  $\geq 0.50$ )  
Dimming centroid offset 0.15 px (3 arcsec) (gate  $\leq 1.0$  px, i.e. on the target)  
Nearest bright-enough neighbour in cutout: TIC 59490529  
4.49 px away, dTmag -0.88 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.

# TIC 469606998

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

## BLS EPHEMERIS

Period 2.714119 d (ALIASED, see below)  
-> corrected period: 5.428238 d  
Epoch (t0) 1739.7341 BTJD  
Duration 0.1200 d  
Depth 3.535 % (35,350 ppm)  
SNR 859.1

## CATALOG CROSS-MATCH (novelty)

|                              |          |
|------------------------------|----------|
| VSX                          | no match |
| ASAS-SN                      | no match |
| Gaia eclipsing-binary table  | no match |
| Gaia variable classification | no match |
| TESS EB catalog (Prsa+2022)  | no match |
| ExoFOP TOI list              | no match |

## INDEPENDENT CHECKS (f3i gates)

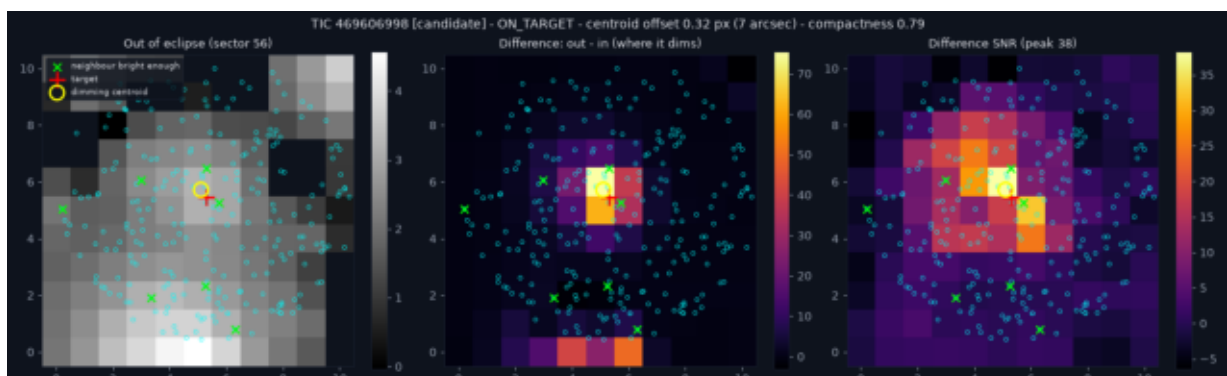
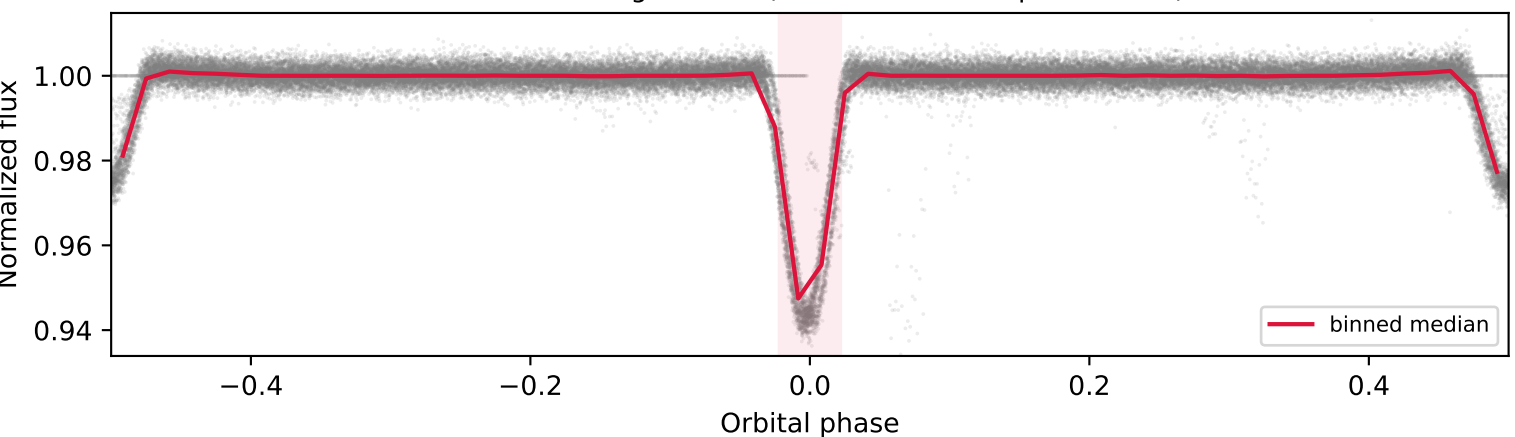
Eclipses observed 70 (gate  $\geq 3$ )  
Robust depth ratio 1.22 (gate  $\geq 0.50$ )  
Robust depth sigma 849.5 (gate  $\geq 7.0$ )  
Odd/even z-score 6.55 (ALIAS FLAGGED)  
Secondary eclipse detected, phase 0.49, 6.1 sigma

Verdict: NOVEL

## PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 56 light-curve dip in this sector: 411.6 sigma t0 refined by 0.0025 phase  
Difference-image peak SNR 38.28 (gate  $\geq 5.0$ ) Compactness 0.79 (gate  $\geq 0.50$ )  
Dimming centroid offset 0.32 px (7 arcsec) (gate  $\leq 1.0$  px, i.e. on the target)  
Nearest bright-enough neighbour in cutout: TIC 469607001  
0.77 px away, dTmag +2.51 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 6.55 (gate  $> 3.0$ ): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 5.428238 d, in any write-up.